UNIT

Two

Complex numbers

- Pre-requirements for unit two : Algebraic form of a complex number.

Exercise **4** Trigonometric form of a complex number.

Exercise **5** Exponential form of a complex number (Euler form).

Exercise **6** De Moivre's theorem.

Exercise **7** Cubic roots of unity.



Exercise

on pre-requirements for unit two
(Algebraic form of a complex number)

From the school book | Understand | Apply | Higher Order Thinking Skills

1 Choose the correct answer from those given :

(1) If $3x + 2yi - 6 + 4xi = 0$, then $(x , y) =$ .....

(a) $(2 , 4)$ (b) $(2 , -4)$ (c) $(-2 , 4)$ (d) $(-2 , -4)$

(2) The S.S. of the equation : $x^2 + 9 = 0$ in C is .....

(a) $\{3 , -3\}$ (b) $\{i , -i\}$ (c) $\{3i , -3i\}$ (d) $\{-9\}$

(3) If $z = 1 + i$, then $\frac{\bar{z}}{z} =$ .....

(a) 2 (b) 1 (c) $-i$ (d) i

(4) If $z = a + bi$, $z + \bar{z} = 6$, then $a =$ .....

(a) 3 (b) -3 (c) 6 (d) -6

(5) If $x + yi = i(2 + i)$, then $x - y =$ .....

(a) 3 (b) -3 (c) 6 (d) -6

(6) If $z = 2 - i$, then $z - \bar{z} =$ .....

(a) zero (b) 4 (c) $-2i$ (d) $2i$

(7) If $a + bi = i(1 - 5i)^2$, then $b =$ .....

(a) 10 (b) -10 (c) 24 (d) -24

(8) $i^n \times i^{n+1} \times i^{n+2} \times i^{n+3} =$ .....

(a) zero (b) 1 (c) -1 (d) i

(9) If L , M are the two roots of the equation : $2ix^2 - 4ix + 3i^3 = 0$, then $L^2 + M^2 =$ .....

(a) 2 (b) 4 (c) 5 (d) 7

(10) $i^{57} + \frac{1}{i^{25}} =$ .....

(a) zero (b) 1 (c) 2 (d) -1

(11) $(1+i)^{12}$ = ...

(a) -8 (b) 32 (c) 64 (d) -64

(12) $(1+i)^5\left(1+\frac{1}{i}\right)^5$ = ...

(a) -8 (b) 32 (c) 64 (d) -64

(13) $i+i^2+i^3+\cdots+i^{100}$ = ...

(a) zero (b) 1 (c) 2 (d) 100

(14) $\left(\frac{2i}{1+i}\right)^5$ = ...

(a) $1+i$ (b) $1-i$ (c) $-4-4i$ (d) $4-4i$

(15) If $\frac{a^2+b^2}{a+bi}=2+3i$, then $a \times b$ = ... when $a, b \in \mathbb{R}^*$

(a) -6 (b) -5 (c) 5 (d) 6

(16) If $2-i$ is a root for the 2nd degree equation whose coefficients are real , then the other root for this equation is .....

(a) -3 (b) $2+i$ (c) $-2+i$ (d) $2-i$

(17) If z_1 , z_2 are two conjugate complex numbers , then $\frac{1}{z_1}+\frac{1}{z_2}$ could be equal to .....

(a) $9-4i$ (b) $5i$ (c) 13 (d) $1+i$

(18) If L , M are the roots of the quadratic equations : $x^2+1=0$, then $L^{2022}+M^{2022}$ = ...

(a) $-2i$ (b) $2i$ (c) -2 (d) 2018

(19) If a , b , c , d are four consecutive integers : $i^a+i^b+i^c+i^d$ = ...

(a) zero (b) -1 (c) 1 (d) i

(20) $i^{|0}+i^{|1}+i^{|2}+i^{|3}+\cdots+i^{|25}$ = ...

(a) zero (b) $2i$ (c) $2i+25$ (d) $2i+20$

2 Find the complex number which equals each of the following :

(1) $(3-2i)^2+(3+2i)^2$

(2) $(1+i)^4-(1-i)^4$

(3) $(1-i^5)(1-i^{10})(1-i^{15})$

(4) $\left(\sqrt{8}-\sqrt{-54}\right)\left(\sqrt{18}-\sqrt{-24}\right)$

(5) $\left(2-\frac{1}{i}\right)^6$

(6) $\frac{3}{2+i}$

(7) $\frac{2+i}{3-4i}$

(8) $\frac{4-5i}{7i}$

3 **Find S.S. for each of the following in C :**

(1) $x^2 + 1 = 0$

(2) $x^4 - 16 = 0$

(3) $x^2 + 4x + 5 = 0$

(4) $x^2 - 6x + 25 = 0$

4 **Find the real value of x , y which satisfy each of the following equations :**

(1) $3x - 2y\,i = (5 - i)^2$

(2) $x^2 - y^2 + i(x + y) = 3(1 + i)$

(3) $(x + 2y\,i)^2 + \frac{8(1 - i)}{1 + i} + 15 = 0$

5 If (-3) is a root of the equation : $x^3 + x^2 - x + 15 = 0$, then find the other two roots.

6 If $z \in \mathbb{C}$ and $z^2 - \bar{z} = 0$ find the value of z and prove that there are two values for the number z are conjugates.

7 If z is a complex number , find the solution set of the equation : $2z - 3\bar{z} = 5 + 10i$

Exercise 4

On trigonometric form of a complex number

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1 **Write the trigonometric and algebraic forms for each of the following complex numbers :**

(1) The number whose modulus = 2 , its amplitude = 0

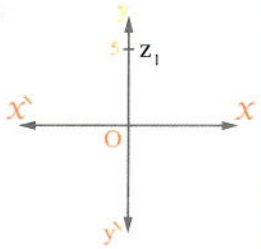
(2) The number whose modulus = 2 , its amplitude = $\frac{3\pi}{2}$

(3) The number whose modulus = 1 , its amplitude = 210°

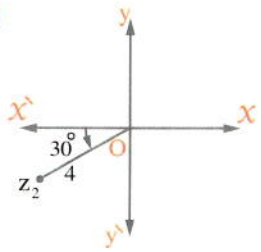
(4) The number whose modulus = $\sqrt{2}$, its amplitude = $\frac{5\pi}{4}$

2 **Write each of the following complex numbers in their trigonometric form :**

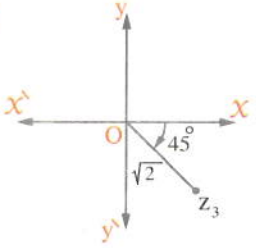
(1)



(2)



(3)



3 **Find the modulus and the principle amplitude for each of the following complex numbers , then put each of them in the trigonometric form :**

(1) -3 | (2) $2i$ | (3) $1+i$

(4) $-\sqrt{3}+i$ | (5) $-1-i$ | (6) $4-4\sqrt{3}\,i$

4 **Put each of the following numbers in the form $x + y\,i$ where $x , y \in \mathbb{R}$, then write their trigonometric form :**

(1) $\frac{8}{\sqrt{3}+i}$ | (2) $\frac{2-2i}{2i}$ | (3) $\frac{5-\sqrt{3}\,i}{\sqrt{3}-2i}$

5 **Choose the correct answer from those given :**

(1) The number $z = 3 - 4i$ is represented on Argand's diagram by the point A where A = ...

(a) (3 , 4) (b) (3 , – 4) (c) (– 3 , 4) (d) (– 3 , – 4)

(2) If the point A $(\sqrt{3}, -1)$ represents a complex number on Argand's diagram , then the argument and the principle amplitude for the number z are .....

(a) $(2, \frac{\pi}{6})$ (b) $(2, \frac{5\pi}{6})$ (c) $(2, \frac{-5\pi}{6})$ (d) $(2, \frac{-\pi}{6})$

(3) ***(2nd Session 2023)*** If the principle amplitude of the complex number z is $(\frac{\pi}{6})$ and its modulus is 1 length unit , then the complex number z in the algebraic form is .....

(a) $\frac{\sqrt{3}}{2} - \frac{1}{2}$ i (b) $\frac{\sqrt{3}}{2} + \frac{1}{2}$ i (c) $\frac{-\sqrt{3}}{2} + \frac{1}{2}$ i (d) $\frac{-\sqrt{3}}{2} - \frac{1}{2}$ i

(4) If $z = -1 + \sqrt{3}$ i then $|\bar{z}| =$ .....

(a) $-1 - \sqrt{3}$ i (b) $\sqrt{2}$ (c) 2 (d) $2\sqrt{2}$

(5) ***(1st Session 2023)*** If the conjugate of the complex number z is $\bar{z} = -1 + \sqrt{3}$ i , then the trigonometric form of the complex number z is .....

(a) $4\left(\cos\left(\frac{-2\pi}{3}\right) + i \sin\left(\frac{-2\pi}{3}\right)\right)$ (b) $2\left(\cos\left(\frac{-2\pi}{3}\right) + i \sin\left(\frac{-2\pi}{3}\right)\right)$

(c) $2\left(\cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3}\right)$ (d) $4\left(\cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3}\right)$

(6) The complex number $z = -2$ i in trigonometric form equals .....

(a) $2(\cos 90° + i \sin 90°)$ (b) $2(\cos -90° + i \sin -90°)$

(c) $2(\cos 0° + i \sin 0°)$ (d) $2(\cos 180° + i \sin 180°)$

(7) The principle amplitude of the complex number $z = 1 - i$ is .....

(a) $\frac{\pi}{4}$ (b) $\frac{-\pi}{4}$ (c) $\frac{-7\pi}{4}$ (d) $\frac{7\pi}{4}$

(8) 📖 Which of the following represents the algebraic form of $2\left(\cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3}\right)$?

(a) $-\sqrt{3} + i$ (b) $-1 + \sqrt{3}$ i (c) $-\sqrt{2} + \sqrt{2}$ i (d) $-\sqrt{3} - \sqrt{3}$ i

(9) If z is a complex number , its principle amplitude is θ , then

First : Amplitude $(\bar{z})$ = ...

(a) θ (b) $-\theta$ (c) $\frac{\pi}{2} - \theta$ (d) $\pi - \theta$

Second : Amplitude $(2z)$ = ...

(a) θ (b) $-\theta$ (c) 2θ (d) -2θ

Third : Amplitude $(\frac{1}{z})$ = ...

(a) θ (b) $-\theta$ (c) $\pi - \theta$ (d) $-\pi + \theta$

(10) If $z = \frac{1}{\bar{z}}$, then $|z| =$ .....

(a) zero (b) 1 (c) $\frac{1}{2}$ (d) ± 1

(11) If $|z| = 10$, then $z\,\bar{z} =$ .....

(a) 10 (b) 100 (c) 1 (d) -100

(12) If $|z| = 6$, then $|-\bar{z}| =$ .....

(a) 6 (b) -6 (c) $\frac{1}{6}$ (d) $-\frac{1}{6}$

(13) If $z = 6\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$, then $|\bar{z}| =$ .....

(a) 6 (b) -6 (c) $3\sqrt{3}$ (d) $6\sqrt{3}$

(14) *(1st Session 2024)* If $z = 6\left(\cos\left(\frac{-5\pi}{4}\right) + i\sin\left(\frac{-5\pi}{4}\right)\right)$, then the principle amplitude of the complex number z = ...

(a) $\frac{3\pi}{4}$ (b) $\frac{\pi}{4}$ (c) $\frac{-\pi}{4}$ (d) $\frac{-3\pi}{4}$

(15) *(1st Session 2017)* The principle amplitude for the number $2\left[\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right]$ is .....

(a) $\frac{\pi}{4}$ (b) $\frac{-\pi}{4}$ (c) $\frac{3\pi}{4}$ (d) $\frac{-3\pi}{4}$

(16) If $|z| + |\bar{z}| = 12$, then $|iz| =$ .....

(a) 12 (b) 12 i (c) 6 (d) -6

(17) If z is a non-zero complex number , then principle amplitude of (z) + principle amplitude of $(\bar{z})$ = ...

(a) zero (b) 90° (c) 180° (d) $-90°$

(18) If the amplitude for the number $z = 28°$, then the amplitude for the number $\left(\frac{i}{\bar{z}}\right)$ = ...

(a) 28° (b) 62° (c) 118° (d) $-28°$

(19) *(Speciman 2025)* If $\frac{2+3z}{z} = 4 + i$ where z is complex number , $i^2 = -1$, then the argument of the complex number z is .....

(a) $\frac{3\pi}{4}$ (b) $\frac{\pi}{4}$ (c) $\frac{-\pi}{4}$ (d) $\frac{-3\pi}{4}$

(20) If $z + 2 = i(z - 2)$, then the complex number z = ...

(in the trigonometric form)

(a) $2(\cos 0° + i\sin 0°)$ (b) $2(\cos 90° + i\sin 90°)$

(c) $2(\cos 180° + i\sin 180°)$ (d) $2(\cos -90° + i\sin -90°)$

(21) *(1st Session 2018)* If $z = \frac{2-i}{2+i}$, then $|z| =$ .....

(a) 3 (b) 4 (c) 1 (d) 5

(22) *(2nd Session 2024)* If the amplitude of the complex number z is $\frac{\pi}{3}$, then the amplitude of the complex number z^3 is .....

(a) $\frac{\pi}{9}$ (b) $\frac{\pi}{6}$ (c) π (d) 2π

(23) If $x = \cos 17° + i \sin 17°$, $y = \cos 11° + i \sin 11°$, then complex number $z = x^3 y^9$ in the trigonometric form is .....

(a) $\cos 30° + i \sin 30°$ (b) $\cos 150° + i \sin 150°$

(c) $\cos -120° + i \sin -120°$ (d) $\cos -60° + i \sin -60°$

(24) *(2nd Session 2023)* If $z = \frac{2+4i}{2-i}$, then the trigonometric form of complex number $z^6 =$ ..... where $i^2 = -1$

(a) $64\left(\cos\frac{\pi}{2} + i \sin\frac{\pi}{2}\right)$ (b) $64(\cos\pi + i \sin\pi)$

(c) $32(\cos\pi + i \sin\pi)$ (d) $32\left(\cos\left(\frac{-\pi}{2}\right) + i \sin\left(\frac{-\pi}{2}\right)\right)$

(25) The additive inverse of (i) represent on Argand diagram by the point .....

(a) (1 , 0) (b) (– 1 , 0) (c) (0 , 1) (d) (0 , – 1)

(26) If $z_1 = 2\sqrt{3} - i$, $z_2 = 2(\cos 30° + i \sin 30°)$, then $\overline{z_1} - \overline{z_2} =$ .....

(a) $2\sqrt{3} + i$ (b) $\sqrt{3} - i$ (c) $\sqrt{3} + 2i$ (d) $3\sqrt{3} + 2i$

(27) If principle argument of z_1 is θ_1 , argument of z_2 is θ_2 , then argument of $z_1 z_2$ is .....

(a) $\theta_1 + \theta_2$ (b) $\theta_1 \times \theta_2$ (c) $\theta_1 - \theta_2$ (d) $\pi - (\theta_1 + \theta_2)$

(28) *(1st Session 2024)* If $z_1 = 5(\cos 30° + i \sin 30°)$, $z_2 = 4(\cos 60° + i \sin 60°)$, then modulus of $z_1 z_2 =$ .....

(a) 20 (b) 10 (c) 9 (d) 8

(29) *(Speciman 2025)* If $z_1 = -2 + 2\sqrt{3}\,i$, $z_2 = 2(\sin 60° + i \cos 60°)$, then the principle amplitude of $\left(\frac{z_2}{z_1}\right) =$ .....

(a) $\frac{5\pi}{3}$ (b) $\frac{\pi}{3}$ (c) $\frac{-\pi}{3}$ (d) $\frac{-\pi}{2}$

(30) If $z_1 = \cos\theta + i \sin\theta$, $z_2 = \cos\theta - i \sin\theta$, then the argument of $\left(\frac{z_1^3}{z_2^4}\right) =$ .....

(a) 7θ (b) θ (c) $-\theta$ (d) -7θ

(31) ***(Speciman 2025)*** If $z_1 = 1 + \sqrt{3}\,i$, $z_2 = \sqrt{2}\,(\cos 45° + i \sin 45°)$, then $\left(\frac{z_1}{z_2}\right)^6 =$ .....

(a) 8 (b) 8 i (c) 8 + 8 i (d) 8 – 8 i

(32) The trigonometric form for the number $(i^{25})^3$ where $i^2 = -1$ is .....

(a) $\cos\frac{\pi}{2} + i \sin\frac{\pi}{2}$ (b) $\cos\pi + i \sin\pi$

(c) $\cos(-\pi) + i \sin(-\pi)$ (d) $\cos\left(\frac{-\pi}{2}\right) + i \sin\left(\frac{-\pi}{2}\right)$

(33) ***(1st Session 2022)*** If z is a complex number where $z = i^{22} + i^{24n-13}$, then the trigonometric form of the number z could be equal to .....

(where $n \in \mathbb{Z}^+$, $i^2 = -1$)

(a) $\sqrt{2}\left(\cos\left(\frac{-\pi}{4}\right) + i \sin\left(\frac{-\pi}{4}\right)\right)$ (b) $\sqrt{2}\left(\cos\frac{\pi}{4} + i \sin\frac{\pi}{4}\right)$

(c) $\sqrt{2}\left(\cos\frac{3\pi}{4} + i \sin\frac{3\pi}{4}\right)$ (d) $\sqrt{2}\left(\cos\left(\frac{-3\pi}{4}\right) + i \sin\left(\frac{-3\pi}{4}\right)\right)$

(34) ***(2nd Session 2022)*** If z is a complex number where $z = i - 6$, then the trigonometric form of the complex number $z + 5$ is ..... , where $i^2 = -1$

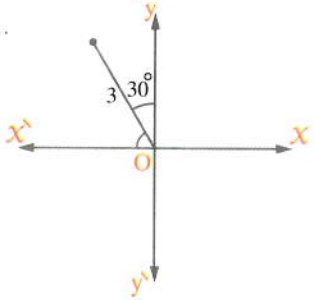
(a) $\sqrt{2}\left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right)\right)$ (b) $\sqrt{2}\left(\cos\left(\frac{-\pi}{4}\right) + i \sin\left(\frac{-\pi}{4}\right)\right)$

(c) $\sqrt{2}\left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right)\right)$ (d) $\sqrt{2}\left(\cos\left(\frac{-3\pi}{4}\right) + i \sin\left(\frac{-3\pi}{4}\right)\right)$

(35) ***(2nd Session 2022)*** If $z = 3\left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right)\right)$, then $\frac{1}{\bar{z}} =$ .....

(a) 9 z (b) $\frac{1}{3}$ z (c) 3 z (d) $\frac{1}{9}$ z

(36) The opposite figure represents the complex number .....



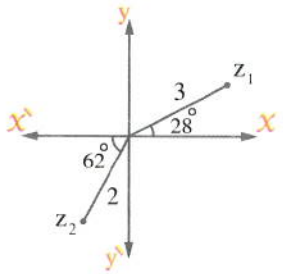
(a) 3 (cos 30° + i sin 30°)

(b) 3 (cos 60° + i sin 60°)

(c) 3 (cos 120° + i sin 120°)

(d) 3 (cos 150° + i sin 150°)

(37) The opposite figure represents two complex numbers z_1 , z_2 , then the amplitude of $(z_1\, z_2)$ = ...



(a) $\frac{\pi}{2}$ (b) $\frac{-\pi}{2}$

(c) π (d) $\frac{-\pi}{8}$

(38) If z is a non-real complex number , which of the following numbers have same amplitude z ?

(a) – z (b) $\bar{z}$ (c) $\frac{1}{z}$ (d) $\frac{1}{\bar{z}}$

(39) If $z = -3(\cos 45° + i \sin 45°)$, then the amplitude of z = ...

(a) $-135°$ (b) $135°$ (c) $45°$ (d) $-45°$

(40) *(2nd Session 2024)* If $z = k\left(\sin\frac{\pi}{4} + i\cos\frac{\pi}{4}\right)$ where $k \in \mathbb{R}^-$, then the principle amplitude of the complex number z equals .....

(a) $\frac{-3\pi}{4}$ (b) $\frac{-\pi}{4}$ (c) $\frac{\pi}{4}$ (d) $\frac{3\pi}{4}$

(41) If the principle amplitude of the number z_1 is $\frac{4\pi}{5}$ and the principle amplitude of z_2 is $\frac{2\pi}{3}$, then the principle amplitude of $\left(\frac{z_1}{z_2}\right)$ = ...

(a) $\frac{2\pi}{15}$ (b) $\frac{\pi}{15}$ (c) $\frac{22\pi}{15}$ (d) $\frac{3\pi}{10}$

(42) If the number z is pure imaginary and has a principle amplitude is $(4\theta + 10°)$, then θ = ...

(a) $20°$ (b) 20 or $-25°$ (c) $90°$ or $-90°$ (d) $-90°$

(43) If z_1 , z_2 are two conjugate numbers and the amplitude of $z_1 = (3\theta - 110°)$ and the amplitude of $z_2 = (2\theta + 50°)$, then θ = ...

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{8}$ (c) $\frac{\pi}{12}$ (d) $\frac{\pi}{15}$

(44) If the modulus of a complex number z is 2 and its amplitude is $\frac{\pi}{2}$, then .....

First : Modulus of the complex number (z + 2) = ...

(a) 2 (b) $\sqrt{5}$ (c) $\sqrt{6}$ (d) $2\sqrt{2}$

Second : Principle amplitude of the complex number (z + 2) = ...

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $\frac{3\pi}{4}$ (d) π

(45) If z is a complex number. Which of the following statements are false ?

(a) $z\bar{z} = |z|^2$ (b) $z\bar{z} = |\bar{z}|^2$

(c) Amplitude (z) = Amplitude $(\bar{z})$ (d) $\overline{z_1 + z_2} = \bar{z_1} + \bar{z_2}$

(46) If z is a complex number , then which of the following complex numbers has the same amplitude as z ?

(a) z + 1 (b) z + i (c) 2 z (d) $-2z$

(47) If $z_1 = 10(\cos\theta + i\sin\theta)$, $z_2 = 5(\cos\alpha + i\sin\alpha)$ and if $\theta - \alpha = \frac{3\pi}{4}$, then $\frac{z_1}{z_2}$ = ...

(a) $-\sqrt{2} + \sqrt{2}\,i$ (b) $\sqrt{2} - \sqrt{2}\,i$ (c) $1 - \sqrt{3}\,i$ (d) $-1 + \sqrt{3}\,i$

(48) If $z = \cos\theta - i\sin\theta$, then $\frac{z^2 - 1}{z\,i} =$ .....

(a) $-2\sin\theta$ (b) $2\sin\theta$ (c) $\sin\theta$ (d) $\sin 2\theta$

(49) If $z_1 = \cos\theta_1 + i\sin\theta_1$ where $0 < \theta_1 < \frac{\pi}{2}$, $z_2 = \cos\theta_2 + i\sin\theta_2$, where $0 < \theta_2 < \frac{\pi}{2}$ and if $z_1^{-4}\, z_2^3 = 1$, then $\frac{\theta_1}{\theta_2} =$ .....

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{7}{8}$ (d) $\frac{2}{3}$

(50) *(1st Session 2021)* If $z_1 = \cos\theta + i\sin\theta$, $z_2 = \cos 2\theta + i\sin 2\theta$, then the principle amplitude of the complex number $3\, z_1\, z_2 =$ ..... , where $\theta \in]0 , \frac{\pi}{6}[$

(a) 9θ (b) θ (c) 5θ (d) 3θ

(51) *(1st Session 2021)* If $z = k\left(\sin\frac{4}{3}\pi - i\cos\frac{4}{3}\pi\right)$, then $z^6 =$ ..... , where $k >$ zero

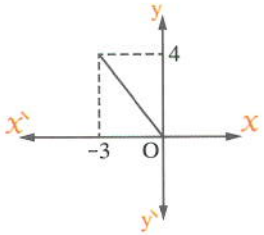
(a) k^6 (b) $6k$ (c) $-k^6$ (d) $-6k$

(52) If the point A represents the complex number z on Argand's plane and point B represents the number $\bar{z}$ on Argand's plane , then B is the image of A by reflection in .....

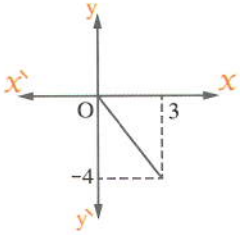
(a) the origin point. (b) the $\mathcal{X}$-axis.

(c) they y-axis. (d) the straight line $y = \mathcal{X}$.

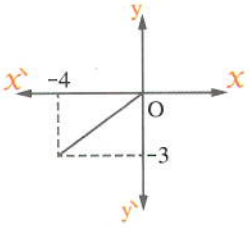
(53) *(1st Session 2025)* If $z = 3 + 4i$, is a complex number , then the figure which represents the complex number $(-\bar{z})$ is .....



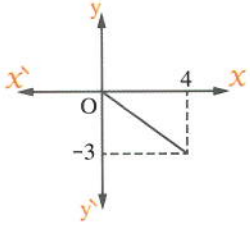
(a)



(b)

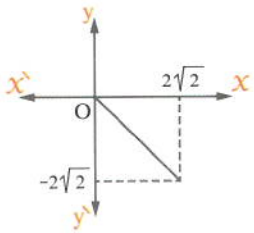


(c)

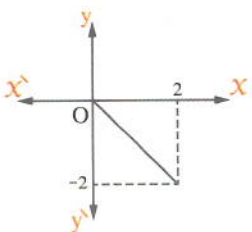


(d)

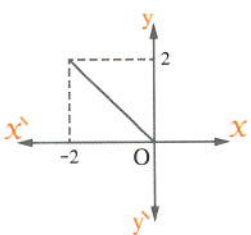
(54) *(Speciman 2025)* Which of the following is the graphical representation of the complex number $z = 2\sqrt{2}\left(\cos\left(\frac{-\pi}{4}\right) + i\sin\left(\frac{-\pi}{4}\right)\right)$?



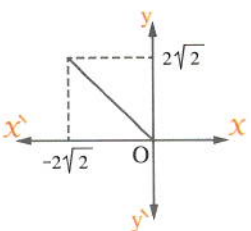
(a)



(b)



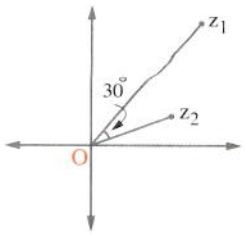
(c)



(d)

(55) ***(Speciman 2025)*** **In the opposite figure :**

z_1 , z_2 are two complex numbers , if $|z_1| = 4$, $|z_2| = 2$, then $\frac{z_1^2}{z_2^2}$ = ...



(a) $2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

(b) $4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

(c) $2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

(d) $4\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

6 If $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$, find the trigonometric form for each of the following numbers : $-z$, $\bar{z}$, $\frac{1}{z}$, $\frac{1}{\bar{z}}$

7 If $z = -\sqrt{3} + i$, find the trigonometric form for each of the following numbers :

z , $-z$, $\bar{z}$, $\frac{1}{z}$, $\frac{1}{\bar{z}}$

8 **Find the modulus and the amplitude of each of the following numbers where θ is acute angle :**

(1) $-8(\cos 45° + i\sin 45°)$

(2) $4(\cos 150° - i\sin 150°)$

(3) $\sin 240° + i\cos 240°$

(4) $-2(\sin 210° - i\cos 210°)$

(5) $2\left(\sin\frac{4}{3}\pi - i\cos\frac{4}{3}\pi\right)$

(6) $1 + i\tan 20°$

(7) $r(\cos\theta - i\sin\theta)$

(8) $-r(\cos\theta + i\sin\theta)$

(9) $r(-\cos\theta + i\sin\theta)$

(10) $r(\sin\theta - i\cos\theta)$

9 **Find the number z in the trigonometric form where $i^2 = -1, \theta \in]-\frac{\pi}{2}, \frac{\pi}{2}[$ in each of the following :**

(1) $z = \frac{1}{1 + i\tan\theta}$

(2) $z = \frac{1 + i\tan\theta}{1 - i\tan\theta}$

(3) $z = \frac{\sin\theta - i\cos\theta}{\sin\frac{\theta}{2} + i\cos\frac{\theta}{2}}$

10 **Find the product in the trigonometric form in each of the following :**

(1) $z_1 = r_1(\cos(a+b) + i\sin(a+b))$, $z_2 = r_2(\cos(a-b) + i\sin(a-b))$

(2) $z_1 = 12(\cos 120° + i\sin 120°)$, $z_2 = \frac{3}{4}(\cos 150° - i\sin 150°)$

(3) $z_1 = -2\left(\cos\frac{3\pi}{4} - i\sin\frac{3\pi}{4}\right)$, $z_2 = 8\left(\sin\frac{\pi}{4} + i\cos\frac{\pi}{4}\right)$

(4) $z_1 = -\sin 240° - i\cos 240°$, $z_2 = 7(\sin 120° - i\cos 120°)$

11 **Find the quotient in the trigonometric form in each of the following :**

(1) $z_1 = 3(\cos 3\theta - i \sin 3\theta)$, $z_2 = 4(\cos 4\theta - i \sin 4\theta)$

(2) $z_1 = r(\cos \theta - i \sin \theta)^2$, $z_2 = r(\cos \theta + i \sin \theta)$

(3) $z_1 = 1$, $z_2 = (\cos 2\theta + i \sin 2\theta)^2$

(4) $z_1 = 10(\cos 300° - i \sin 300°)$, $z_2 = 2(\sin 300° - i \cos 300°)$

(5) $z_1 = 14(\cos 15° + i \sin 15°)^2$, $z_2 = 7(-\sin 75° - i \cos 75°)^2$

12 **Express each of the following in the form $x + y\,i$:**

(1) $3\left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12}\right) \times 4\left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12}\right)$

(2) $3\left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12}\right) \times \sqrt{2}\left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}\right)$

(3) $\dfrac{\sqrt{2}\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)}{\frac{1}{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)}$

13 🕮 **If z_1 , z_2 are two complex numbers on Argand's plane in the two given figures find $\frac{z_2}{z_1}$ in the form $x + y\,i$:**

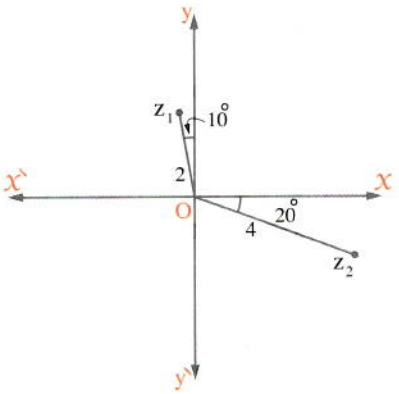


Figure (1)

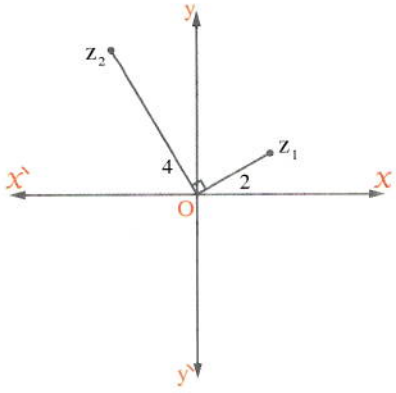


Figure (2)

14 🕮 If $z_1 = 2(\cos 10° + i \sin 10°)$, $z_2 = 3(\cos 40° + i \sin 40°)$

Find the number $z_1^4 \, z_2^2$ in the form $x + i\,y$

15 If $z_1 = 1 - \sqrt{3}\,i$, $z_2 = 1 + i$

Find each of the following in trigonometric form :

(1) 🕮 $z_1 \, z_2$ (2) 🕮 $\frac{z_2}{z_1}$ (3) $(z_1)^8$ (4) 🕮 $(z_2)^6$

16 If $z_1 = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$, $z_2 = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6}$

Find the modulus and the amplitude of the number $\left(\frac{z_1}{z_2}\right)^n$ where $i^2 = -1$

17 If $z = \frac{(1+i)(2-i)}{(1-i)(3+i)}$ Find $|z|$

18 If $z = \frac{(a+b)+i(a-b)}{(a-b)-i(a+b)}$

Find the number z in the simplest form , then find $|z|$ where a , $b \in \mathbb{R}$

19 *(2nd Session 2010)* Put the number $z_1 = \left(\cos\frac{4\pi}{3} - i\sin\frac{4\pi}{3}\right)$, $z_2 = \sqrt{3} - i$

in trigonometric form , then find the modulus and the principle amplitude of the number z where $z = z_1 z_2$

20 *(2nd Session 2006)* If $z = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$ where $i^2 = -1$, find each of the two numbers : $z_1 = z - 1$, $z_2 = z + 1$ in the trigonometric form , then prove that the complex number $\frac{z_1}{z_2}$ is a pure imaginary number.

21 If z_1 , z_2 are two complex numbers where $z_1 = -9 + 3\sqrt{3}\,i$

$|z_2| = \sqrt{3}$, amplitude $z_2 = \frac{7\pi}{12}$, find each of the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $-\pi < \theta \leq \pi$

(1) z_1 **(2)** z_2 **(3)** $z_1 z_2$ **(4)** $\frac{z_1}{z_2}$

22 If $z_1 = \frac{(1+\sqrt{3}\,i)(1-i)}{1+i}$, $z_2 = \frac{(1-\sqrt{3}\,i)(1+i)}{1-i}$

Find $\frac{z_1}{z_2}$ in the trigonometric form.

23 If $z_1 = \cos 114° + i\sin 66°$, $z_2 = \cos 42° + i\sin 138°$

$z_3 = \sin 24 + i\sin 114$, find the algebraic form of the number : $\frac{z_1 z_2}{z_3}$

24 **Choose the correct answer from those given :**

(1) Multiplicative inverse of the complex number z where $|z| \neq 1$ is .....

(a) $\bar{z}$ (b) $\frac{\bar{z}}{|z|^2}$ (c) $\frac{\bar{z}}{|z|}$ (d) $\frac{z}{|z|^2}$

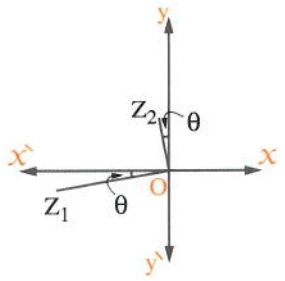
(2) The complex number $\left(\frac{1+2i}{1-i}\right)$ lies in Argand's plane in the ..... quadrant.

(a) 1st (b) 2nd (c) 3rd (d) 4th

(3) *(Speciman 2025)* If θ is the principle amplitude of the complex number $z = 3 - 7i$, then $\tan 2\theta =$ .....

(a) $\frac{14}{3}$ (b) $\frac{3}{14}$ (c) $\frac{20}{21}$ (d) $\frac{21}{20}$

(4) *(2nd Session 2022)* The opposite figure represents two complex numbers z_1 , z_2 on argand's plane. If $|z_1| = 2|z_2|$, then the trigonometric form of the complex numer z where $z = \frac{z_2}{z_1}$ is .....



(a) $\frac{1}{2}\left(\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)\right)$

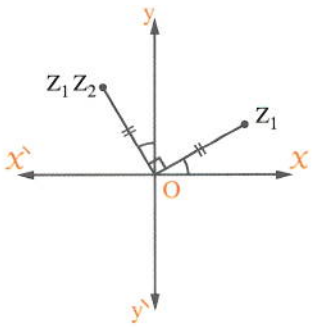
(b) $2\left(\cos\left(\frac{-\pi}{2}\right) + i\sin\left(\frac{-\pi}{2}\right)\right)$

(c) $\frac{1}{2}\left(\cos\left(\frac{-\pi}{2}\right) + i\sin\left(\frac{-\pi}{2}\right)\right)$

(d) $2\left(\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)\right)$

(5) **In the opposite figure :**

z_1 , z_2 are two complex numbers and $(z_1\ z_2)$ is a complex number , then z_2 = ...

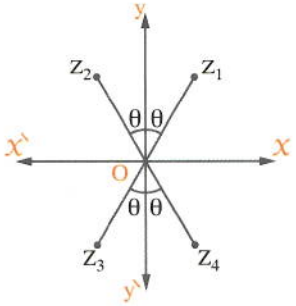


(a) – z i (b) – i

(c) i (d) z i

(6) **In the opposite figure :**

If z_1 , z_2 , z_3 and z_4 are complex numbers , then arg $(z_1\ z_2\ z_3\ z_4)$ = ...



(a) – 90° (b) zero

(c) 90° (d) 180°

(7) ABCD is a square drawn on Argand's diagram its centre is the origin , the point A represents the complex number $\left(1 + \sqrt{3}\,i\right)$, then the point C represents the complex number .....

(a) $-1 - \sqrt{3}\,i$ (b) $1 - \sqrt{3}\,i$ (c) $-1 + \sqrt{3}\,i$ (d) $2\,(1 + \sqrt{3}\,i)$

(8) If z_1 , z_2 are two complex numbers , $\arg(z_1 z_2) = \frac{5\pi}{18}$, $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{9}$, then $\arg(z_1) =$ .....

(a) $\frac{5\pi}{36}$ (b) $\frac{7\pi}{36}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{4}$

(9) If $\arg(z_1 z_2) = \frac{\pi}{6}$, $\arg(z_1 z_3) = \frac{2\pi}{9}$, $\arg(z_2 z_3) = \frac{5\pi}{18}$, then $\arg(z_1 z_2 z_3) =$ .....

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{5}$ (d) $\frac{\pi}{6}$

(10) If z is a complex number , and $\arg(z-2) = \frac{\pi}{2}$, $\arg(z-4) = \frac{3\pi}{4}$, then $\arg(z) =$ .....

(a) π (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) $\frac{3\pi}{4}$

(11) If $z_1 = 3 + 3\sqrt{3}\,i$, $z_2 = -4 - 4\sqrt{3}\,i$, then the amplitude of $(z_1 + z_2)$ = ...

(a) $60°$ (b) $-120°$ (c) $180°$ (d) $-60°$

(12) If $z = 1 + \sqrt{3}\,i$, then the principle amplitude of the number $\left(1 + \sqrt{3}\,i\right)^8$ is .....

(a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{-2\pi}{3}$

(13) *(1st Session 2019)* If $z = \left(1 + \sqrt{3}\,i\right)^n$ and $|z| = 8$, then the principle amplitude of z is .....

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) π

(14) If $z_1 = 2i$, $z_2 = -1 + 3i$, where $i^2 = -1$, then $\arg(z_1 - z_2)$ equals .....

(a) $\frac{3\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $-\frac{\pi}{4}$ (d) $-\frac{3\pi}{4}$

(15) If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ and if $\theta_1 + \theta_2 = \pi$, then $z_1 z_2$ = ...

(a) $r_1 r_2$ (b) $-r_1 r_2$ (c) $r_1 r_2 i$ (d) $-r_1 r_2 i$

(16) If the argument of the complex number $z \in]-\pi, 0[$, then $\arg(-z) - \arg(z)$ = ...

(a) π (b) $-\pi$ (c) $-\frac{\pi}{2}$ (d) $\frac{\pi}{2}$

(17) If z_1 , z_2 are two non-zero complex numbers and $|z_1| = |z_2|$ and $\arg(z_1) + \arg(z_2) = \pi$, then z_1 = ...

(a) z_2 (b) $\overline{z_2}$ (c) $-z_2$ (d) $-\overline{z_2}$

(18) If $|z_1| = |z_2|$ and amplitude of (z_1) + amplitude of $(z_2) = 0$, then .....

(a) $\overline{z_1} = z_2$ (b) $z_1 = z_2$ (c) $z_1 + z_2 = 0$ (d) $\overline{z_1} = \overline{z_2}$

(19) If z is a complex number whose argument θ , $|z| = 1$, then the argument of $\left(\frac{1+z}{1+\bar{z}}\right)$ equals .....

(a) $-\theta$ (b) $\frac{\pi}{2} - \theta$ (c) θ (d) $\pi - \theta$

(20) If $|z| = |z - 2|$, then the real part of the number z equals .....

(a) 1 (b) -1 (c) -2 (d) 2

(21) If z is a complex number and $|z - 3| = |z|$ and the principle amplitude for the number (z) equals $\frac{-\pi}{4}$, then $|z| =$ .....

(a) $3\sqrt{2}$ (b) $\frac{3\sqrt{2}}{2}$ (c) $2\sqrt{2}$ (d) 2

(22) If $z = x + y\,i$ is a complex number and $\left|\frac{z - 3\,i}{z + 3\,i}\right| = 1$, then z lies ..... on Argand's plane

(a) on the x-axis (b) on the y-axis

(c) in the first quadrant (d) in the second quadrant

(23) If $z = \frac{i}{\cos 75^\circ + i \sin 75^\circ}$, then the principle amplitude for the number (z^6) equals .....

(a) $\frac{5\pi}{6}$ (b) $\frac{2\pi}{3}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

(24) If the principle amplitude for the number $(\bar{z}) = \frac{-\pi}{3}$, then the principle amplitude for the number $(z + i\,z)$ equals ..... where z is a complex number.

(a) $\frac{\pi}{3}$ (b) $\frac{5\pi}{12}$ (c) $\frac{7\pi}{12}$ (d) $\frac{3\pi}{4}$

(25) If the principle amplitude for the complex number (z^4) more than three times the principle amplitude for the number (5 z) by $\frac{\pi}{3}$, then the principle amplitude for the number $(\bar{z})$ equals .....

(a) $\frac{\pi}{3}$ (b) $\frac{-\pi}{3}$ (c) $\frac{-\pi}{33}$ (d) $\frac{2\pi}{3}$

(26) If z is a complex number and $z = \sin 20^\circ\,(1 + i \cot 20^\circ)$, then

First : $|z| =$ .....

(a) 1 (b) $\sin 20^\circ$ (c) $\cos 20^\circ$ (d) $\tan 20^\circ$

Second : The principle amplitude for the number (z) equals .....

(a) 20° (b) 40° (c) 50° (d) 70°

(27) The modulus of the complex number $z = (1 + i \tan 15^\circ)$ equals .....

(a) $\sin 15^\circ$ (b) $\cos 15^\circ$ (c) $\tan 15^\circ$ (d) $\sec 15^\circ$

(28) If z is a complex number where $z + |z| = 8 + 12i$, then $|z|^2 =$ .....

(a) 121 (b) 144 (c) 169 (d) 228

(29) If $\frac{\cos\theta - i\sin\theta}{\cos 3\theta - i\sin 3\theta} = i$, then $\theta =$ .....

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{5\pi}{12}$

(30) If $x + yi = \frac{a+bi}{a-bi}$, then $x^2 + y^2 =$ .....

(a) $a^2 + b^2$ (b) $a^2 - b^2$ (c) $-ab$ (d) 1

(31) If a is a real number , then the conjugate of the number $\frac{a^3+i}{a^2+ai-1}$ is .....

(a) $a - i$ (b) $a + i$ (c) $\frac{a^3-i}{a^2+ai-1}$ (d) $\frac{a^3+i}{a^2-ai-1}$

(32) If $z = \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^n$ where n is a positive integer , $z = 1$, then the least value of n is = ...

(a) 9 (b) 6 (c) 3 (d) 1

(33) Amplitude of the complex number $[(1 - \cos\theta) + i\sin\theta]$ equals ..... , where $0 < \theta < \pi$

(a) $\frac{\pi}{4} - \frac{\theta}{2}$ (b) $\frac{\pi}{2} - \frac{\theta}{2}$ (c) $\frac{\pi}{2} - \theta$ (d) $\frac{\pi}{2} - \frac{\theta}{4}$

(34) If $z = 1 + \cos 2\theta + i\sin 2\theta$, then $|z| =$ ..... where θ is an acute angle.

(a) 1 (b) 2 (c) $2\cos\theta$ (d) $2\sin 2\theta$

(35) If $z = 1 + \cos 40° + i\sin 40°$, then arg (z) = ...

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{9}$ (c) $\frac{\pi}{18}$ (d) $\frac{2\pi}{9}$

(36) ***(Speciman 2025)*** If the number $z = \left(\sqrt{3} - \cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)^6$, then z = ...
(where $i^2 = -1$)

(a) $3\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ (b) $\cos\pi + i\sin\pi$

(c) $27(\cos\pi + i\sin\pi)$ (d) $3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

(37) ***(Speciman 2025)*** $\left(\frac{1 + \cos\frac{\pi}{12} + i\sin\frac{\pi}{12}}{1 + \cos\frac{\pi}{12} - i\sin\frac{\pi}{12}}\right)^{72} =$ .....

(a) zero (b) -1 (c) 1 (d) $\frac{-1}{2}$

(38) If $z_1 = \cos 70° + i\sin 70°$, $z_2 = \cos 20° + i\sin 20°$, then

First : arg. $(z_1 + z_2)$ = ...

(a) 90° (b) 50° (c) 45° (d) 140°

Second : The modulus of the number $(z_1 + z_2)$ = ...

(a) $\sqrt{2}$ (b) $\cos 20°$

(c) $\sin 20°$ (d) $\sqrt{2}\ (\cos 20° + \sin 20°)$

(39) If $z_1 = \cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3}$, $z_2 = \cos\frac{\pi}{6} + i \sin\frac{\pi}{6}$, then the amplitude of $\left(\frac{z_1}{z_2}\right)^7$ is .....

(a) $\frac{\pi}{6}$ (b) π (c) $\frac{-1}{2}\pi$ (d) 2π

(40) The amplitude of the complex number $z = \frac{1 + i \tan 18°}{1 - i \tan 18°}$ is .....

(a) $\frac{1}{2}\pi$ (b) $\frac{1}{4}\pi$ (c) $\frac{1}{5}\pi$ (d) $\frac{1}{12}\pi$

(41) If $x - \frac{1}{x} = i$, then $x^{32} + x^{-32}$ = ...

(a) -1 (b) Zero (c) 1 (d) 2

(42) If z_1 , z_2 are two non-zero complex numbers and $|z_1 + z_2| = |z_1| + |z_2|$, then amplitude (z_1) – amplitude (z_2) = ...

(a) π (b) $\frac{\pi}{2}$ (c) $-\pi$ (d) zero

(43) ***(2nd Session 2019)*** If $A = \cos\theta + i \sin\theta$, $B = \cos\alpha + i \sin\alpha$, then the value of $\frac{1}{2}\left(AB + \frac{1}{AB}\right)$ equals .....

(a) $\sin(\theta + \alpha)$ (b) $\sin(\theta - \alpha)$ (c) $\cos(\theta + \alpha)$ (d) $\cos(\theta - \alpha)$

(44) $\frac{(\cos\theta + i \sin\theta)^4}{(\sin\theta + i \cos\theta)^5}$ = ...

(a) $\sin 9\theta - i \cos 9\theta$ (b) $\cos 9\theta - i \sin 9\theta$

(c) $\cos\theta - i \sin\theta$ (d) $\sin\theta - i \cos\theta$

(45) If the amplitude for the complex number $(z - 1 + i)$ is π and if $|z| = 2$, then z = ...

(a) $1 - \sqrt{3}\,i$ (b) $-1 + \sqrt{3}\,i$ (c) $-\sqrt{3} - i$ (d) $3 - i$

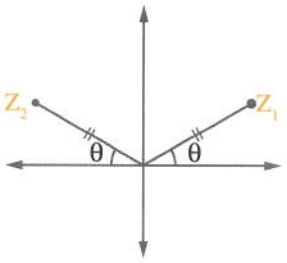
(46) In the opposite figure :

If $|z_1| = |z_2| = r$

, then z_2 = ...

(a) $\overline{z_1}$ (b) $-z_1$

(c) $-\overline{z_1}$ (d) $z_1\, i$



(47) If z_1 , z_2 are two complex numbers such that $\overline{z_1} + i\,\overline{z_2}$ = zero , arg $(z_1 z_2) = \pi$, then arg (z_1) = ...

(a) $\frac{-3\pi}{4}$ (b) $\frac{3\pi}{4}$ (c) $\frac{3\pi}{2}$ (d) $\frac{-3\pi}{2}$

(48) If z is a complex number , then the equation : $|z| = 3$ represents on Argand diagram by .....

(a) a circle , its centre at the origin and its raduis is 3

(b) a circle , its centre is the point (0 , 3) and its raduis is 3

(c) a circle , its centre is the point (3 , 0) and its raduis is 3

(d) a circle , its centre is the origin and its raduis is 9

(49) If z is a complex number , then the equation $|z|^2 + 8(z + \bar{z}) + z\bar{z} = 8$ represents a circle , its surface area = ... area units.

(a) 9π (b) 16π (c) 20π (d) 25π

(50) If the points A , B , C represent on an Argand's plane the complex numbers z , $-z$, $\bar{z}$ respectively , where $z = 5(\cos\theta + i\sin\theta)$, θ is an acute angle , where $\sin\theta = \frac{3}{5}$, then the area of the triangle ABC = ... area unit.

(a) 5 (b) 10 (c) 24 (d) 25

(51) If $z_1 = 1 - \sqrt{3}\,i$, $|z_1 z_2| = 1$, $\arg(z_2) - \arg(z_1) = \frac{\pi}{2}$, then area of the triangle whose vertices z_1 , z_2 , O (the origin) equals .....

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) 2

(52) If the point (A) represents on Argand's plane the complex number (z) and the point (B) represents the complex number (i z) on the same plane , then $m(\angle AOB)$ = ... , where (O) is origin point.

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) π

(53) If ABCD is a quadrilateral and if $z_1 = 3(\cos A + i\sin A)$, $z_2 = 2(\cos B + i\sin B)$, $z_3 = 4(\cos C + i\sin C)$, $z_4 = \cos D + i\sin D$, then $z_1 z_2 z_3 z_4$ = ...

(a) 48 (b) 24 (c) 24 i (d) 48 i

---

25 If z is a complex number , $|z| = z + 3 - 2i$, then find z

---

26 If $\arg(z + i) = \frac{\pi}{4}$, $\arg(z - 3) = \frac{3\pi}{4}$, then find z in the algebraic form where z is a complex number.

27 If $\arg z_1 = \frac{\pi}{3}$, $\arg z_2 = \frac{3\pi}{4}$, $\arg z_3 = \frac{\pi}{6}$, **find :**

(1) $\arg\left(z_1^3 z_2^2\right)$ **(2)** $\arg(2 z_1 . z_2)$

(3) $\arg\left(\frac{z_1 z_2}{z_3}\right)$ **(4)** $\arg\left(z_3^6\right)$

28 If $|z_1| = |z_2| = 1$, and the arg $(z_1 z_2^3) = 81°$, arg $(\frac{z_1}{z_2}) = 33°$,
write in the form of $x + y\,i$ the number $(z_1^{15} . z_2^{15})$

29 If $|z_1| = |z_2| = 1$, arg $(z_1 \times z_2^5)$ equals $105°$, arg $(z_1 \div z_2)$ equals $33°$ and
if $x + y\,i = z_1^{14} + z_2^{15}$, then find value of each x , y « -1 , -1 »

30 Simplify to the simplest form : $\dfrac{(\cos\frac{\pi}{9} + i\sin\frac{\pi}{9})^3 (\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})^4}{(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3})^4}$ « $\frac{1}{2} + \frac{\sqrt{3}}{2}\,i$ »

31 **Prove that :** $\dfrac{(1+i)^5 \times (\cos\theta + i\sin\theta)}{(-1+i)^4} = \sqrt{2}\left[\cos\left((\frac{\pi}{4} + \theta\right) + i\sin\left(\frac{\pi}{4} + \theta\right)\right]$

32 If $z_1 = \cos 75° + i\sin 75°$, $z_2 = \cos 15° + i\sin 15°$,
find the trigonometric form of the number : $z_1 + z_2$ « $\sqrt{3}\,(\cos 45° + i\sin 45°)$ »

33 If $z = \cos\theta + i\sin\theta$, $z_1 = \dfrac{1+z}{1-z}$, prove that : $z_1 = i\cot\dfrac{\theta}{2}$

34 If $\theta \in]-\pi , \pi]$, find the modulus and amplitude of the number :
$z = 1 + \cos\theta + i\sin\theta$

35 If $x + \dfrac{1}{x} = 2\cos\theta$, prove that : $x^n - \dfrac{1}{x^n} = \pm 2\,i\sin(n\,\theta)$

36 If $z_1 = 4(\cos 2\theta + i\sin 2\theta)$, $z_2 = 13(\cos\theta - i\sin\theta)$ where $\theta \in]0 , \frac{\pi}{2}[$, $\tan\theta = \frac{5}{12}$
Find the trigonometric form and the algebraic form of each $z_1 z_2$, z_2^2

37 If $z = 5(\cos\theta + i\sin\theta)$ where $\theta \in]\frac{3\pi}{2} , 2\pi[$, $\cos\theta = \frac{3}{5}$,
find the trigonometric form and the algebraic form of each z^{-1} , z^{-2} , $(\bar{z})^{-2}$

38 **Prove that :** $\left(\dfrac{\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}}{\sin\frac{\theta}{2} - i\cos\frac{\theta}{2}}\right)^n = \cos n\left(\frac{\pi}{2} - \theta\right) + i\sin n\left(\frac{\pi}{2} - \theta\right)$

39 ***(2nd Session 2002)*** By putting $\varnothing = \dfrac{\pi}{2} - 2\theta$ or otherwise , $i^2 = -1$, prove that :
$\left(\dfrac{1 + \sin\varnothing + i\cos\varnothing}{1 + \sin\varnothing - i\cos\varnothing}\right)^n = \cos n\left(\frac{\pi}{2} - \varnothing\right) + i\sin n\left(\frac{\pi}{2} - \varnothing\right)$

40 **Prove that :** $i\left((1-i)^n - (1+i)^n\right) = 2^{\frac{n+2}{2}}\sin\dfrac{n\pi}{4}$

41 If $z_1 \in \mathbb{C}$, $z_2 \in \mathbb{C}$, prove that : $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left(|z_1|^2 + |z_2|^2\right)$

42 If $x + y\,i = (a + b\,i)^n$, prove that : $x^2 + y^2 = (a^2 + b^2)^n$

43 Find using the complex numbers the value : $\frac{\sin 5x}{\sin x}$ in terms of $\cos x$

44 Find $(1 + i)^n$ in each of the two forms , the algebraic and trigonometric and then deduce the values of : a , b if $a + b\,i = (1 + i)^n$

Higher skills

45 **Choose the correct answer from those given :**

(1) If z_1 , z_2 , z_3 are three complex numbers such that :

$|z_1| = |z_2| = |z_3| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = 1$ and $|z_1 + z_2 + z_3| = a$, then .....

(a) $a = 1$ (b) $a < 1$ (c) $a > 3$ (d) $a = 3$

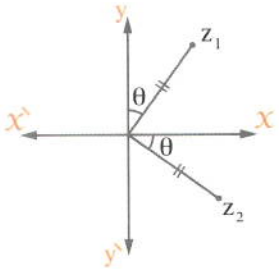
(2) The points represents the complex numbers z , z i , z + i z in the Argand's plane are vertices of Δ ABC , then the area of Δ ABC = ...

(a) $|z|$ (b) $|z|^2$ (c) $2|z|^2$ (d) $\frac{1}{2}|z|^2$

(3) **In the opposite figure :**

If $|z_1| = |z_2| = \sqrt{3}$

, then $|z_1 + z_2|$ = ...



(a) 2 (b) $2\sqrt{3}$

(c) $\sqrt{6}$ (d) 3

(4) If $\left(2\sqrt{2} + i\right)^{50} = 3^{49}(a + b\,i)$ where a , $b \in \mathbb{R}$, then $a^2 + b^2$ = ...

(a) 9 (b) 25 (c) 49 (d) 50

(5) If $z_1 = 4\sqrt{3}(\cos 140° + i \sin 140°)$, $z_2 = 4(\cos 230° + i \sin 230°)$

, then $|z_1 - z_2|$ = ...

(a) 8 (b) $8\sqrt{2}$ (c) 12 (d) $12\sqrt{2}$

(6) If z is a complex number and $|z - 4| < |z - 2|$, then the real part of the complex number could be equal.....

(a) 1 (b) 2 (c) 3 (d) 4

(7) If $z_n = \cos\frac{\pi}{2^n} + i \sin\frac{\pi}{2^n}$, then $(z_1 \times z_2 \times z_3 \times \ldots \text{ to } \infty)$ equals .....

(a) – 1 (b) zero (c) 2 (d) ∞

Exercise 5

On exponential form of a complex number (Euler form)

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **Put in the exponential form (Euler form) each of the following :**

(1) $-2i$

(2) 7

(3) $2\sqrt{3}-2i$

(4) $\dfrac{4}{1+\sqrt{3}\,i}$

2 **Find the algebraic form of the number z in each of the following :**

(1) $z = 2e^{\pi i}$

(2) $z = \sqrt{2}\,e^{\frac{-\pi}{2}i}$

(3) $z = 8e^{\frac{\pi}{6}i}$

3 **Express each of the following in the exponential form :**

(1) $\frac{1}{2}(\cos 60° + i \sin 60°)$

(2) $-4\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right)$

(3) $-2\left(\sin\frac{\pi}{3} - i\cos\frac{\pi}{3}\right)$

(4) $\sqrt{2}\left(-\sin\frac{3\pi}{4} + i\cos\frac{3\pi}{4}\right)$

4 **If $z = \sqrt{6} - 3\sqrt{2}\,i$, find the trigonometric form and the exponential form of each :**

(1) z

(2) $\bar{z}$

(3) $-z$

(4) $\frac{1}{\bar{z}}$

5 **If $\theta \in]0, \frac{\pi}{2}[$, write each of the following complex numbers in exponential form :**

(1) $r(\cos\theta - i\sin\theta)$

(2) $-r(\cos\theta + i\sin\theta)$

(3) $6(-\cos\theta + i\sin\theta)$

(4) $2(-\sin\theta + i\cos\theta)$

(5) $-r(\sin\theta + i\cos\theta)$

(6) $\dfrac{1}{\cos\theta - i\sin\theta}$

6 **Put the number z in each of the following in form $x + y\,i$, then find the trigonometric form and the exponential form of each :**

(1) $z = \dfrac{2-2i}{2i}$

(2) $z = \dfrac{\sqrt{2}\,i}{1+i}$

(3) $z = \dfrac{(i-1)(3-4i)}{(1+2i)^2}$

7 **Choose the correct answer from those given :**

(1) $e^{2\pi i} =$ ..... (in the algebraic form)

(a) $1+i$ (b) $1-i$ (c) 1 (d) -1

(2) $3\,i$ = ... (in the exponential form)

(a) $e^{-\frac{\pi}{2}i}$ (b) $3\,e^{\frac{\pi}{2}i}$ (c) $3\,e^{\frac{-\pi}{2}i}$ (d) $3\,e^{\pi i}$

(3) The algebraic form of the complex number $z = \sqrt{2}\,e^{\frac{\pi}{4}i}$ is .....

(a) $1 + i$ (b) $1 - i$ (c) $-1 + i$ (d) $-1 - i$

(4) $\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right)^4 = e^{.....}$

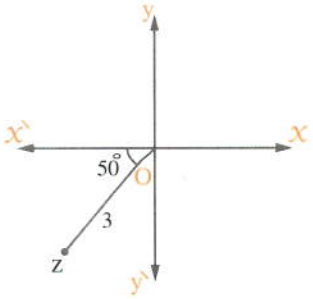
(a) $\frac{2\pi}{3}i$ (b) $-\frac{1}{3}\pi i$ (c) $\frac{1}{3}\pi i$ (d) $\frac{-5}{3}\pi i$

(5) If $z = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$, then $z + 1$ = ... (in the exponential form)

(a) $e^{\frac{\pi}{6}i}$ (b) $\sqrt{3}\,e^{\frac{\pi}{6}i}$ (c) $\sqrt{3}\,e^{\frac{\pi}{3}i}$ (d) $\left(\sqrt{2}+1\right)e^{\frac{\pi}{3}i}$

(6) In the opposite figure :

The number z in the exponential form = ...



(a) $e^{\frac{13}{18}\pi i}$ (b) $e^{-\frac{13}{18}\pi i}$

(c) $3\,e^{\frac{13}{18}\pi i}$ (d) $3\,e^{-\frac{13}{18}\pi i}$

(7) ***(1st Session 2022)*** If $z_1 = 2(\cos 150° - i\sin 150°)$, $z_2 = 3(\sin 150° + i\cos 150°)$, then the exponential form of the number $z_1 z_2$ could be equals .....

(a) $6\,e^{\frac{5}{6}\pi i}$ (b) $6\,e^{-\frac{5}{6}\pi i}$ (c) $6\,e^{\frac{\pi}{3}i}$ (d) $6\,e^{\frac{\pi}{2}i}$

(8) ***(2nd Session 2017)*** If $z = 2 + 2\sqrt{3}\,i$, then the exponential form for z equals .....

(a) $4\,e^{\frac{-\pi}{3}i}$ (b) $4\,e^{\frac{\pi}{3}i}$ (c) $4\,e^{\frac{-\pi}{6}i}$ (d) $4\,e^{\frac{\pi}{6}i}$

(9) ***(Trial 2023)*** If $z_1 = 8(\cos\pi + i\sin\pi)$ and $z_2 = 4\,e^{\frac{3\pi}{2}i}$, then $\frac{z_1}{z_2}$ = ... where $i^2 = -1$

(a) $2 + 2i$ (b) $2i$ (c) $-2i$ (d) $2 - 2i$

(10) ***(2nd Session 2023)*** If z_1 and z_2 are two complex numbers where $z_1 = e^{3+\frac{\pi}{2}i}$, $z_2 = e^{3-\frac{\pi}{2}i}$, then $\frac{z_2}{z_1}$ = ...

(a) i (b) 1 (c) $-i$ (d) -1

(11) ***(1st Session 2024)*** If $z_1 = 1 + \sqrt{3}\,i$, $z_2 = 4\left(\cos\left(\frac{-\pi}{3}\right) + i\sin\left(\frac{-\pi}{3}\right)\right)$, then $\frac{z_2}{z_1}$ = ...

(a) $2\,e^{\frac{2\pi}{3}i}$ (b) $\frac{1}{2}\,e^{\frac{2\pi}{3}i}$ (c) $\frac{1}{2}\,e^{\frac{-2\pi}{3}i}$ (d) $2\,e^{\frac{-2\pi}{3}i}$

(12) *(1st Session 2018)* $e^{\theta i} + e^{-\theta i} =$ .....

(a) $e^{2\theta i}$ (b) $2\cos\theta$ (c) $2\sin\theta$ (d) $e^{-2\theta i}$

(13) *(1st Session 2019)* The numerical value of the expression $e^{\pi i} - e^{-\pi i} =$ .....

(a) -2 (b) 0 (c) 1 (d) 2

(14) If $z = x + i\,y$, then the real part of the number e^{z} is .....

(a) $e^{x}\cos y$ (b) $e^{x}\sin y$ (c) e^{x} (d) $\cos y$

(15) *(Trial 2023)* If $z_1 = e^{\frac{\pi}{3} - \frac{\pi}{3} i}$, and $z_2 = e^{\frac{\pi}{6} - \frac{\pi}{6} i}$ then $z_1\, z_2 =$ ..... where $i^2 = -1$

(a) $e^{\frac{-\pi}{2} i}$ (b) $-e^{\frac{\pi}{2} i}$ (c) $-e^{\frac{\pi}{2} i}$ (d) $e^{\frac{\pi}{2} i}$

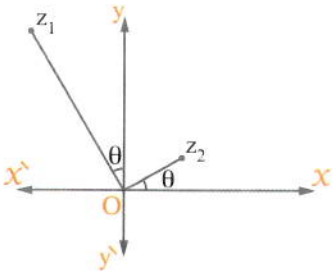
(16) *(Trial 2023)* If $z_1 = 2\,e^{\frac{\pi}{6} i}$, $z_2 = 2\,e^{\frac{-\pi}{6} i}$ and $z = z_1 + z_2 + z_1\, z_2$

, then $z =$ ..... where $i^2 = -1$

(a) $2\sqrt{3} + 1$ (b) $2\sqrt{3}$ (c) $2\left(\sqrt{3} + 2\right)$ (d) $\sqrt{3} + 2$

(17) In the opposite figure :

z_1 , z_2 are two complex numbers in Argand's plane $|z_1| = 3\,|z_2|$ then $\frac{z_1}{z_2} =$ .....



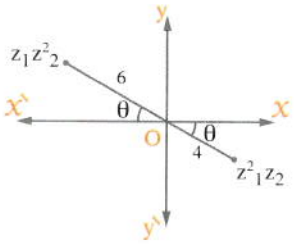
(a) $3\,e^{-\frac{\pi}{2} i}$ (b) $3\,e^{\frac{\pi}{2} i}$

(c) $\frac{1}{3}\,e^{\frac{\pi}{2} i}$ (d) $\frac{1}{3}\,e^{-\frac{\pi}{2} i}$

(18) *(2nd Session 2024)* **In the opposite figure :**

If $(z_1\, z_2^2)$, $(z_1^2\, z_2)$ are two complex numbers represented on Argand's plane

, then $\frac{z_1}{z_2} =$ .....



(a) $\frac{2}{3}\,e^{\frac{-\pi}{2} i}$ (b) $\frac{2}{3}\,e^{\pi i}$ (c) $\frac{3}{2}\,e^{\pi i}$ (d) $\frac{3}{2}\,e^{\frac{-\pi}{2} i}$

(19) If $z_1 = \cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}$ and $|z_2| = 4$, $\arg(z_1\, z_2) = \frac{-2\pi}{3}$, then $z_2 =$ .....

(a) $4\,e^{\frac{2\pi}{7} i}$ (b) $4\,e^{\frac{\pi}{3} i}$ (c) $4\,e^{\frac{\pi}{6} i}$ (d) $4\,e^{\frac{\pi}{12} i}$

(20) If $z_1 = r_1\,e^{\theta i}$, $z_2 = r_2\,e^{\theta i}$, then $|z_1 + z_2|$ .....

(a) less than $(r_1 + r_2)$ (b) greater than $(r_1 + r_2)$

(c) equals $(r_1 + r_2)$ (d) equals $(r_1 - r_2)$

(21) If $z = r\,e^{\theta i}$, then $r\,e^{\left(\frac{\pi}{2}+\theta\right)i}$ represents by the complex number .....

(a) $\frac{\pi}{2}\,z$ (b) $z + \frac{\pi}{2}$ (c) $\frac{z}{i}$ (d) $z\,i$

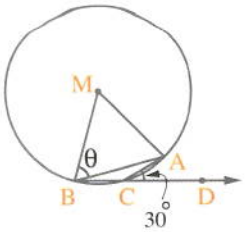
(22) In the opposite figure :

Circle with center (M)

in which m ($\angle$ ACD) = 30°

, m ($\angle$ MBA) = θ and $z = \sin 3\theta + i \cos 3\theta$

, then z = ... in the exponential form



(a) $e^{\frac{3\pi}{4}i}$ (b) $e^{\frac{-3\pi}{4}i}$ (c) $e^{\frac{\pi}{2}i}$ (d) $e^{\frac{-\pi}{2}i}$

(23) *(1st Session 2021)* If z is a complex number and $z + \bar{z} = 2\,e^{\pi i}$, then z may be equal .....

(a) $e^{\pi i}$ (b) $2\,e^{\frac{\pi}{2}i}$ (c) $e^{\frac{-\pi}{2}i}$ (d) $2\,e^{\pi i}$

(24) *(2nd Session 2021)* If z_1 , z_2 are two complex numbers , $z_1 = e^{5+k\pi i}$, $z_2 = e^{(5+k i)\pi}$, where $-\frac{1}{2} < k < \frac{1}{2}$, then the principle amplitude of the complex number $z_1 + z_2$ could be equal .....

(a) $\frac{2\pi}{3}$ (b) $-\frac{\pi}{2}$ (c) π (d) $-\frac{\pi}{6}$

(25) *(2nd Session 2021)* If the point A $\left(\sqrt{k}\,,-\sqrt{k}\right)$ represents the complex number z on the Argand's plane , where $k > 1$, then the exponential form of the number z is .....

(a) $\sqrt{2k}\,e^{-\frac{\pi}{4}i}$ (b) $\sqrt{2}\,k\,e^{-\frac{\pi}{4}i}$ (c) $\sqrt{2k}\,e^{\frac{\pi}{4}i}$ (d) $\sqrt{2}\,k\,e^{\frac{\pi}{4}i}$

(26) *(Speciman 2025)* If $z = \frac{1+\sqrt{3}\,i}{\sqrt{3}+i}$, then the exponential form of the complex number $(\bar{z})^{100}$ is .....

(a) $e^{\frac{-2\pi}{3}i}$ (b) $e^{\frac{-\pi}{3}i}$ (c) $e^{\frac{2\pi}{3}i}$ (d) $e^{\frac{\pi}{3}i}$

(27) *(Speciman 2025)* If $e^{\theta i} = \cos\theta + i \sin\theta$, then $e^{a i} \cdot e^{b i} \cdot e^{c i}$ = ...

where a , b , c are the measures of the angles of a triangle in radian measure. ($i^2 = -1$)

(a) $-i$ (b) -1 (c) 1 (d) i

---

8 **Prove that :**

(1) $e^{\frac{\pi}{3}i} - e^{\frac{-\pi}{3}i} = \sqrt{3}\,i$

(2) $e^{4+\pi i} - e^{-\pi i} = 1 - e^4$

(3) $\cos\theta = \frac{1}{2}\left(e^{\theta i} + e^{-\theta i}\right)$

(4) $\sin\theta = \frac{-i}{2}\left(e^{\theta i} - e^{-\theta i}\right)$

---

9 **Put the complex number $z = \frac{-4}{1-\sqrt{3}\,i}$ in the trigonometric form and the exponential form then :**

Prove that : (1) z^6 is a real number.

(2) $\frac{1}{z} = \frac{1}{2}\,e^{\frac{2\pi}{3}i}$

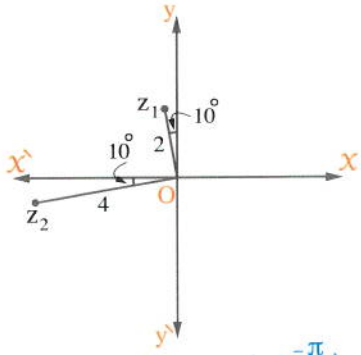
10 If $z_1 = 2(\cos 75° + i \sin 75°)$, $z_2 = 4(\cos 15° + i \sin 15°)$, find in the exponential form the number : $z_1 z_2$, $\frac{z_1}{z_2}$

« $8 e^{\frac{\pi}{2} i}$, $\frac{1}{2} e^{\frac{\pi}{3} i}$ »

11 If $z_1 = \frac{2i}{1+i}$, $z_2 = 4(\cos 150° + \sin 150°)$ find $z_2 \div z_1^2$ in the exponential form. « $2 e^{\frac{\pi}{3} i}$ »

12 **In the opposite figure :**

Find the exponential form of the number $\frac{z_1}{z_2}$



« $\frac{1}{2} e^{\frac{-\pi}{2} i}$ »

13 If $z_1 = 6(\cos 240° + i \sin 240°)$, $z_2 = 2(\cos 300° + i \sin 300°)$

Find the exponential form each of :

(1) $\frac{z_1}{z_2}$ **(2)** $\frac{z_2}{z_1}$ **(3)** z_1^2 **(4)** z_2^4

« $3 e^{\frac{-\pi}{3} i}$, $\frac{1}{3} e^{\frac{\pi}{3} i}$, $36 e^{\frac{2\pi}{3} i}$, $16 e^{\frac{2\pi}{3} i}$ »

14 If $z_1 = 2(\cos 150° - i \sin 150°)$, $z_2 = 3(\sin 150° + i \cos 150°)$

Find the exponential form of each of the following : $z_1 z_2$, z_1^2 , z_2^4

« $6 e^{\frac{5\pi}{6} i}$, $4 e^{\frac{\pi}{3} i}$, $81 e^{\frac{2\pi}{3} i}$ »

15 **Put the result of each of the following in the exponential form and the trigonometric form :**

(1) $2 e^{\frac{\pi}{6} i} \times 3 e^{\frac{2\pi}{3} i}$ **(2)** $4 e^{\frac{7\pi}{4} i} \div 2 e^{\frac{5\pi}{3} i}$

(3) $\frac{3\sqrt{2}\, e^{\frac{7\pi}{15} i} \times \sqrt{2}\, e^{\frac{4\pi}{5} i}}{e^{\frac{\pi}{6} i} \times 2 e^{\frac{2\pi}{3} i}}$

16 ***(1st Session 2006)*** If $z_1 = 4\left(\cos \frac{5\pi}{3} - i \sin \frac{5\pi}{3}\right)$, $z_2 = e^{\frac{\pi i}{2}}$ where $i^2 = -1$, find the number $z = \frac{z_2}{z_1}$ in the trigonometric form.

« $\frac{1}{4}\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$ »

17 ***(2nd Session 2009)*** If $z_1 = 1 + \sqrt{3}\, i$, $z_2 = \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}$, then find : $\frac{z_1}{z_2}$ in the exponential form. « $2 e^{\frac{\pi i}{2}}$ »

18 ***(2nd Session 2003)*** If $z_1 = 1 - \sqrt{3}\, i$ and $z_2 = 1 + i$, $i^2 = -1$, find the number $z = \frac{z_1}{z_2}$ in the exponential form , then prove that z^{12} is a real number.

« $\sqrt{2} e^{\frac{-7\pi}{12} i}$ »

19 *(1st Session 2010)* If $z_1 = \sqrt{2}\left(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right)$, $z_2 = 1 + i$, find the exponential form of the number z where $z = \frac{z_1^2}{z_2}$ « $\sqrt{2}\, e^{\frac{-3\pi}{4}i}$ »

20 *(1st Session 2007)* If $z_1 = 6\left(\cos\frac{4\pi}{3} - i\sin\frac{4\pi}{3}\right)^2$, $z_2 = 2\, e^{\frac{5\pi}{3}i}$, where $i^2 = -1$, find the number $z = z_1 z_2^2$ in the trigonometric form. « $24\left(\cos\frac{2}{3}\pi + i\sin\frac{2}{3}\pi\right)$ »

21 Put each of the two expressions : $\sqrt{2}\, i^5$, $i^4 + i^5$ in the trigonometric form then find the algebraic form and also the exponential form of the expression $\left(\frac{\sqrt{2}\, i^5}{i^4 + i^5}\right)^4$ « -1 , $e^{\pi i}$ »

22 **Put the result of each of the following in the algebraic form :**

(1) $z = 3\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) \times 2\, e^{\frac{\pi}{6}i}$ « $-3\sqrt{3} + 3i$ »

(2) $z = -2(1 - i)\, e^{\frac{\pi}{2}i}$ « $-2 - 2i$ »

23 If $z_1 = \cos 45° + i\sin 45°$, $z_2 = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$, $z_3 = \cos 30° + i\sin 30°$

Find the exponential form (euler form) the expression $\frac{z_1^6 z_2^5}{z_3^4}$ « $e^{\frac{\pi}{2}i}$ »

24 If $z_1 = \cos\frac{7\pi}{4} - i\sin\frac{7\pi}{4}$, $z_2 = -1 + i$, $z_3 = \sin 60° + i\cos 60°$

Find the number : $z = \frac{z_1^4 z_2^2}{z_3^5}$ in the exponential form and algebraic form. « $2\, e^{\frac{-\pi}{3}i}$, $1 - \sqrt{3}\, i$ »

25 If $z = e^{\theta i}$, find modulus and principle amplitude of the number $\frac{1 + z}{1 - z}$ « $\cot\frac{\theta}{2}$, $\frac{\pi}{2}$ »

Higher skills

26 **Choose the correct answer from those given :**

(1) If $a\, e^{2\theta i} + b\, e^{-2\theta i} = 4\cos 2\theta - 6i\sin 2\theta$, then $a \times b =$ ..... where a , b are real numbers.

(a) -1 (b) 5 (c) -5 (d) 4

(2) If $x + y\, i = e^{\theta i}$, then the least value for the number $x\, y$ is .....

(a) $-\frac{1}{2}$ (b) -1 (c) -2 (d) 1

(3) The opposite figure represents two complex numbers z_1 , z_2 on Argand's diagram then $\left(\frac{z_1}{z_2}\right)^6 =$ .....

(a) $8\, e^{\frac{-\pi}{6}i}$ (b) $8\, e^{\frac{-\pi}{4}i}$

(c) $8\, e^{\frac{-\pi}{3}i}$ (d) $8\, e^{\frac{-\pi}{2}i}$

